

Cancer studies in Metabolomics Workbench with exercise, physical-activity, or actigraphy data

Complete corpus re-screen — 4,503 studies, live REST metadata retrieved 2026-08-26. Every claim below was re-derived independently by six parallel verification agents; their corrections are incorporated and attributed. Every `ST` identifier links to its Metabolomics Workbench study page.

Bottom line

The answer depends on what counts, so here it is in tiers. Deposited factors were screened for all 4,503 studies in the corpus; submitter prose was screened for 798 of the 804 cancer studies (99.3%).

Exercise as an intervention: 1 study. [ST002183](#), a randomised trial of individualised exercise in multiple myeloma. The only cancer study where exercise appears in deposited metadata at all — and it appears as a trial-arm value (`Group:exercise`), not as a named exercise variable.

Objective physical function measured: 2 studies. [ST004161](#) (plasma) and [ST004166](#) (muscle) measured handgrip strength, stair-climb power, and CT muscle cross-sectional area in 88 men with gastrointestinal or genitourinary cancer. Those measurements are described in the study prose but are **not deposited** — the factor table carries only weight-loss group.

Rodent voluntary activity measured: 1 study. [ST002163](#) recorded voluntary wheel-running activity in cancer-bearing mice. Again described in prose, not deposited.

Actigraphy, accelerometry, or any wearable-derived measure: 0 studies. Zero in the deposited factors of all 4,503 studies, and zero in the prose of 798 cancer studies.

Lung cancer specifically: 0 studies, at both layers, under all four defensible constructions of the lung set (56, 62, 70, or 75 studies). For lung cancer, exercise-metabolomics is an availability gap in this repository.

The pattern across all of these is the same: where activity or function data exists, it was measured but not deposited. Read every zero as a statement about **what Metabolomics Workbench holds**, not about what the parent studies measured.

Corrections to the first pass

	First pass	Verified	What was wrong
Corpus size	4,499 (snapshot)	4,503 unique studies (4,504 returned rows)	The listing endpoint emits one row per species, so ST002097 appears twice. A corpus size of 4,504 is a row count, not a study count. The +4 versus the snapshot (ST004154 , ST004298 , ST004823 , ST004824) are newly <i>released</i> older submissions, not new accessions — the maximum accession, ST005132, is unchanged.

	First pass	Verified	What was wrong
Cancer studies	478	790 (see range below)	MW publishes a curated Disease field. It labels 770 studies oncologic, of which 312 have no oncology word in their title and were invisible to the first screen — e.g. ST000494 (palmitate isotopomers in LNCaP cells), ST001049 (P4HA1 knockdown in MDA-231), ST000821 (IDH1 mutation profiling). The 478 figure reproduces exactly; it is a recall failure, not an arithmetic one.
Lung-cancer studies	57 (38 human, 19 mouse)	56 to 75 , construction-dependent	The 57 reproduces, but includes a confirmed false positive: ST002058 “Muscle/Lung/Tumor metabolomics” is a 4T1 breast cancer cachexia model. Adding MW’s Disease label and cell-line curation raises the set.
Exposure detector	sample factors only	factors (4,503/4,503) + titles + reverse keyword search + prose	Calibration: of the 149 studies with “exercise” in the title, only 14 encode an exercise term in their factors — ~9% sensitivity . The first pass rested on a detector that misses nine exercise studies in ten.
Search layers	factors + titles	+ reverse keyword search + submitter prose, 798 of 804 cancer studies	The prose sweep found three studies with real activity or physical-function measurements that the factor screen could not see, because the measurements were never deposited as variables. The first pass would have reported those studies as having no activity data.
Wording	“has no exercise variable”	“deposits no exercise variable in MW structured metadata”	MW’s factor field is an arm-and-timepoint slot, not a covariate inventory. The original wording would license a claim the evidence does not support — and which is contradicted for at least one study (below).

The headline changed in one direction only. No second exercise *intervention* exists — four search directions converge on ST002183 for that. But the prose sweep added three studies that measured physical function or voluntary activity without depositing it, which the structured screen alone reported as nothing.

The cancer denominator is a range, and it does not matter

MW’s oncology vocabulary is coarse: the only cancer values in its 258-term Disease field are Cancer (765 studies), Lung cancer (14), and Multiple myeloma (1), plus Cachexia (28). There is no NSCLC, no lung adenocarcinoma, no

mesothelioma, no per-site term at all. So the cancer set has to be assembled, and the total depends on documented judgment calls:

Construction	Studies
Oncology title regex alone	473–478 (regex-dependent)
MW curated Disease field alone, cachexia excluded	770
Union of the two — the figure used throughout	790
Union including the 16 cachexia-only studies	806
Union including cancer-derived cell lines used as generic in-vitro models	827

A hard recall ceiling sits under all of these: **1,775 studies (39.4%) have no Disease annotation at all**, and 1,737 of them have no cancer word in the title either. The true cancer count is bounded below by 790 and is not determinable from these endpoints.

This ambiguity does not affect the answer. The exposure screen was run over the sample factors of **all 4,503 studies in the corpus**, not over a cancer subset. Whichever way the cancer set is drawn, the exercise result is the same, because every candidate was screened.

How the screen was run

1. Structured metadata, whole corpus. Sample-level factors were fetched for all 4,503 studies. Every distinct factor string and sample-source value was matched against an exercise vocabulary: exercise, actigraphy, accelerometer, physical activity, treadmill, wheel running, VO₂max, cardiorespiratory, MVPA, sedentary, step count, pedometer, activity monitor, wearable, aerobic/endurance/resistance/strength training, detraining, bed rest, immobilisation, HIIT, sprint interval, cycle ergometer, walk test, fitness test, athlete, trained/untrained. Thirty-five studies matched corpus-wide; every match was adjudicated by hand.

2. Reverse keyword search. MW’s keyword endpoint was queried for 20 exercise terms and the returned IDs intersected with the cancer set, to catch any study naming an exercise exposure in its title but not its factors.

Term	Hits	Term	Hits	Term	Hits
exercise	149	endurance	117	prehabilitation	0
training	122	fitness	2	rehabilitation	0
physical activity	0	VO2	0	running	3
actigraphy	0	sedentary	1	resistance training	0
accelerometer	0	walking	0	cardiorespiratory	0
aerobic	8	athlete	2	wheel running	0
treadmill	0	cachexia	27		

All 20 queries were reachable, so each zero is a true zero-hit result rather than an unreachable index. The exercise side unions to **193** studies.

3. Submitter prose. Free text is reachable only by an indirect route: the study-level `mwtab` endpoint returns several concatenated JSON documents (one per analysis) and cannot be parsed as one document, so prose must be pulled per analysis via `/rest/study/analysis_id/<AN id>/mwtab`. The design description lives in `STUDY_SUMMARY`, `PROJECT_SUMMARY`, and — where an intervention arm is described — `TREATMENT_SUMMARY`. A verification agent read the prose of 122 cancer studies this way and found exactly one genuine exercise study, ST002183; seven of its eight regex hits were false positives on “training set” and “trained interviewer”.

That blind spot has since been closed. Prose was retrieved and screened for **798 of the 804 cancer studies (99.3%)**, including **74 of the 75 lung studies**. Six studies could not be parsed because of the concatenated-JSON defect described below and are recorded as unavailable, not as negative: [ST001269](#), [ST002967](#), [ST003617](#), [ST003921](#), [ST004438](#), [ST004470](#). Seven studies tripped the exercise/function vocabulary; all seven were adjudicated by hand and are set out in the next section.

Result 1 — the one cancer exercise study

ST002183 — Individualized exercise intervention for people with multiple myeloma improves quality of life in a randomized controlled trial

Field	Value
MW Disease	Multiple myeloma
Species	Homo sapiens
Sample source	Blood (plasma)
Institute	QIMR Berghofer Medical Research Institute
Platform	LC-MS
Rows in factors table	180
Released	2024-02-05
Licence	CC BY 4.0
Factor keys	Group, Time_point — two keys only
Group values	waitlist 75 · exercise 51 · MS2 32 · QC 16 · Blank 6

This is the only exercise signal in MW oncology, and it is thinner than a first read suggests. Five things a human must settle before anyone computes a pre/post contrast from it:

- The exposure is an arm label, not a variable.** The factor keys are the generic `Group` and `Time_point`; the word “exercise” appears only as a value. No dose, modality, session adherence, or objective activity measure is deposited. A trial that coded its arms `A/B` or `INT/CON` would be invisible to this screen entirely.
- The waitlist arm crosses over to the same exercise programme.** It is therefore not an unexposed control at later timepoints, and cannot be treated as one.
- Timepoint coding is asymmetric between arms** — exercise carries `–`, `1`, `3`; waitlist carries `–`, `1`, `2`, `4`. `Time_point:–` is baseline, not a missing value.
- `Group:MS2` and `Group:QC` are analytical injections, not study arms** (54 of the 180 rows), and the factor vocabulary does not mark them as such.
- There is no subject identifier field.** Participant identity is recoverable only by string-parsing sample IDs, and the two analyses’ sample-factor blocks are byte-identical and carry only HILIC-negative filenames, which contradicts the factors endpoint. Quantitative values are not retrievable from the standard named-metabolite REST outputs.

Result 2 — what the prose found that the factors could not

Seven of 798 cancer studies mention an activity, exercise, or physical-function term in their submitter prose. Four are genuine measurements, and none of the four deposits the measurement as a variable.

Study	Species, n	What was measured	Deposited?
ST002183	human, 180 rows	Randomised exercise vs waitlist; prose adds that cardiorespiratory fitness improved	Arm label only (Group:exercise)
ST004161	human, 88 plasma	Handgrip strength, stair-climb power, CT lumbar muscle cross-sectional area, >5% weight loss	No — factors carry only Group:WeightLoss/WeightStable
ST004166	human, 79 muscle	Same three function measures, rectus abdominis biopsy at surgery	No — same two-value factor table
ST002163	mouse, 37 liver	Voluntary wheel-running activity in 4T1 tumour-bearing and sham mice, with and without Nnmt deletion	No — factors carry genotype, tumour group, sex, treatment

The remaining three are not activity data:

- [ST002167](#) is the AML12 cell-culture companion to ST002163 and shares its abstract; the wheel-running sentence describes the mouse study, not this one.
- [ST003315](#) (neoadjuvant breast cancer trial) used ECOG performance status ≤ 2 as an *eligibility criterion*. An entry filter is not a measured variable.
- [ST001237](#) (PD-1 blockade, 1,221 samples) deposits CRF_MSKCC_Risk_Group, a composite prognostic score that incorporates performance status. Functional status is embedded in a risk category, not deposited separately and not recoverable from it.

Why this matters more than the headline count

[ST004161](#) and [ST004166](#) are a paired plasma-and-muscle study in 88 men with cancer, with objective physical-function phenotyping (handgrip strength, stair-climb power) and CT muscularity, explicitly designed to ask which metabolic signals track poor physical function. That is the closest existing analogue in MW oncology to an exercise-metabolomics design. The function measurements exist; they are simply not in the repository. Obtaining them is a data request to the submitting group (VA Puget Sound Health Care System), not a new study.

Note the distinction that has to be kept: handgrip strength and stair-climb power are measures of physical **capacity**, and voluntary wheel running is a measure of **activity**. Neither is an exercise intervention, and neither can be relabelled as one.

Result 3 — no wearable or actigraphy data, anywhere

Across all 4,503 studies, **zero** deposit an actigraphy, accelerometer, wearable, pedometer, step-count, MVPA, IPAQ, or GPAQ variable. The prose sweep reproduces the zero at the second layer: no such term appears in the submitter prose of any of the 798 cancer studies screened, and a verification agent found the same zero independently over its own 122-study prose sample.

This is not a schema limitation. MW can carry a physical-activity variable: [ST001749](#) (REACH Metabolomics Study, Alzheimer's disease) deposits an explicit physical activity factor. No oncology study does.

Result 4 — what cancer studies do deposit about lifestyle

The most informative finding of the verification pass is a selective-deposition pattern. Scanning factor **keys** across the 789-study cancer set:

Subject-level covariate	Cancer studies depositing it
Smoking (Smoker, Smoking Status, pack-years)	11 of 789 (1.4%) — and 8 of 42 human lung studies (19%)
Diet	15 of 789 (1.9%), of which 14 are mouse dietary-intervention arms
BMI or anthropometry	0 of 789 (0.0%)
Physical activity or fitness	0 of 789 (0.0%)

Human lung-cancer studies routinely deposit smoking status — the covariate their analysis needs. They deposit no activity and no BMI. That is a pattern of *selective deposition*, and it is the reason the absence of activity data in MW cannot be read as absence of measurement upstream.

Lung cancer in detail

The lung set has four defensible constructions. **All four screen to zero.**

Construction	Studies	Exercise/activity variables
Lung title regex within the title-cancer set (first pass, minus the ST002058 false positive)	56 (38 human, 18 mouse)	0
Union with MW Disease = Lung cancer	62–64 (42–44 human, 19–20 mouse)	0
Plus cell-line curation (A549, H1299, PC9, 393P)	70 (51 human, 19 mouse)	0
Plus a Lung sample-source annotation within the cancer set	75 (53 human, 22 mouse)	0

Three further facts sharpen the picture:

- **The lung factor vocabulary contains no activity construct at all.** Across the whole set there are 58 distinct factor keys and not one is an activity, fitness, or functional-capacity measure — no ECOG, no Karnofsky, no 6-minute walk. An over-inclusive 44-alternative screen returned only homonyms (`meth` inside `methylation_rank` and `methionine_condition`), and its positive control fired correctly on ST002183.
- **Study count overstates the evidence base.** [ST003883](#)–[ST003893](#) is eleven studies from one paper, and accounts for 10 of the 14 studies MW labels Lung cancer. The 75 studies represent roughly 8 independent mouse investigations plus the human sets.
- **MW records zero mesothelioma studies corpus-wide**, so that branch of any lung screen is dead.

The lung result is now checked at both layers: 0 of 75 in deposited factors, and 0 of the 74 lung studies whose submitter prose could be parsed. Only [ST001269](#) (exosomal lipids in NSCLC staging) remains unscreened in prose, and it is recorded as unavailable.

Why the data is not there

Design archetype	Studies (curated set of 70)
In-vitro / cell line	31
Mouse or rodent in-vivo tumour model	13
Human diagnostic / biomarker discovery	18
Human treatment-response	6
Human prospective / cohort	2

Forty-four of 70 (63%) are cell culture or mouse — designs that cannot host an activity exposure. Of the 26 human studies, 24 are cross-sectional case-control diagnostic or treatment-response designs. Only two are prospective. The lung set's problem is not a missing field; it is that the designs cannot carry one.

The two lung studies that are worth a data request

ST002773 — the strong case. A nested case-control of untargeted plasma metabolomics and lung cancer among never-smoking women, inside the prospective Shanghai Women's Health Study. Primary publication: [PMID 38651675](#), [10.1002/ijc.34929](#), *International Journal of Cancer*, 2024, peer-reviewed, accession-verified.

MW deposits two factor keys for it — `Subject_ID` and `LungCancer` — across 1,152 rows. But MW's own `COLLECTION_SUMMARY` states that the SWHS parent cohort followed 74,942 women “through multiple in-person interviews and self-administered questionnaires to obtain information on demographics, occupational and environmental exposures, lifestyle, dietary, and other factors, including environmental tobacco smoke and body mass index.” The SWHS physical-activity questionnaire is separately published and validated ([PMID 14630608](#)), with MET-hour analyses in the literature ([PMID 17478434](#)).

For this study, the missing activity variable is a **deposition gap, not a measurement gap**. Whether physical activity is available for these specific 790 women is an open data-request question. It must not be scored as a cohort that did not measure activity.

ST002082 — the weaker case. A prospective study of metabolic change in the dying process in lung cancer patients. Primary publication: [PMID 40016594](#), [10.1038/s43856-025-00764-3](#), *Communications Medicine*, 2025, peer-reviewed. MW deposits a single factor key, `Time before death (weeks)`, over 112 urine samples. Its `COLLECTION_SUMMARY` notes that “an anonymised record of the medication administered was collected” — a covariate collected but not deposited, which by itself proves the deposited factor set is a subset of what was recorded. Verdict: **unresolved**, with no positive evidence either way.

Abstract-text scans of both primary publications and two companion papers found no physical-activity, actigraphy, accelerometry, or fitness term (an initial substring scan reported “ECOG” in both; on inspection both were the letters inside “Recognizing”, and were discarded). Abstract silence is not evidence of non-collection, and this is labelled publication-text inference throughout.

What this means for an exercise-oncology aim

1. **A lung-cancer exercise-metabolomics question cannot be answered from Metabolomics Workbench.** No study deposits the exposure, under any construction of the lung set. Represent this as an availability gap; do not approximate it with a nearby cohort.
2. **ST002183 is the only cancer exercise study, and it is multiple myeloma** with a crossover control arm and no quantified exposure. It can support a hypothesis-generating group contrast after the five issues above are adjudicated. It cannot support a lung-cancer claim.
3. **Three data requests, not a dead end.** [ST002773](#) (Shanghai Women's Health Study — validated physical-activity questionnaire upstream, plasma metabolomics deposited) and the paired [ST004161/ST004166](#) (objective handgrip and stair-climb power in 88 men with cancer, plasma and muscle metabolomics deposited) are the highest-value follow-ups in the whole screen. In all three the phenotype was measured and is simply not in the repository.
4. **Do not write the sentence “lung-cancer cohorts do not measure physical activity.”** It is unsupported by this evidence and directly contradicted for ST002773. The defensible sentence is that no lung-cancer study *deposits* an activity variable in Metabolomics Workbench.
5. **For a rodent arm, ST002163 is the only cancer study in MW that measured voluntary activity** (wheel running in 4T1 tumour-bearing mice, with an Nnmt-knockout rescue). The activity data is not deposited, and MW holds only 4 rat cancer studies in total, which limits any MoTrPAC rat-side alignment.

6. **The cachexia cluster is the realistic adjacent resource** for muscle-metabolism questions in cancer — 28 studies under MW’s Cachexia label, including a 12-part human muscle-and-serum series ([ST001005–ST001018](#)) and a 10-tissue mouse series ([ST002881–ST002904](#)). Cachexia is a wasting phenotype, not an activity exposure, and must be labelled as such.

Limits, and what a reader may not conclude

Limit	Consequence
1,775 studies (39.4%) carry no Disease annotation; 1,737 have no cancer word in the title either	The 790-study cancer set is a floor. An unannotated oncology study with a mechanistic title sits outside it.
Factor-only exposure detection is ~9% sensitive	Structured metadata alone cannot close this question, which is why titles, keyword search, and prose were added.
“Human” in the species field does not mean human subjects	389 of 790 cancer studies (49.2%) have in-vitro titles, and MW labels a HeLa or LNCaP study <i>Homo sapiens</i> . No count of “human cancer studies” here should be read as a count of human cohorts. This is text inference.
Prose unavailable for 6 of 804 cancer studies	ST001269 , ST002967 , ST003617 , ST003921 , ST004438 , ST004470 could not be parsed. They are an unscreened residue, not a negative result.
Prose is a submitter-written abstract of variable depth	A study that measured activity and described it in neither its factors nor its abstract is still invisible. The three studies found here were found <i>because</i> their abstracts happened to name the measure.
MW prose is a submitter-written abstract	It is not a data-availability statement. Nothing here resolves what any parent study collected.
Absence is deposition-level throughout	Every zero in this report means “not deposited in Metabolomics Workbench”. For at least one study the parent cohort demonstrably did measure more than it deposited.

Repository defects found along the way

These are FAIR findings in their own right and were reported by the verification agents:

- The study-level `mwtab` endpoint returns **concatenated JSON documents** (one per analysis) with no array wrapper, so it cannot be parsed as one document. Use `/rest/study/analysis_id/<AN id>/mwtab`.
- `/rest/study/study_id/<ST id>/untarg_studies` **ignores the study ID** and returns a corpus-wide listing of 2,395 analyses; two different study IDs return byte-identical payloads. Any per-study fact read from it is a mirage.
- [ST003117](#) is indexed in the disease table (Malaria) but its summary endpoint returns `[]`. [ST003478](#) returns `[]` for both species and disease — unavailable, not absent.
- MW Disease curation is internally inconsistent within a single submission series: of four studies from one lab’s NCI-H1299 MAT2A experiment ([ST004220](#), [ST004221](#), [ST004225](#), [ST004253](#)), three are labelled Lung cancer and one Cancer.
- Europe PMC’s fielded `FULL_TEXT:"ST002773"` query returns zero hits while the unfielded query returns two. A fielded zero there is an index-coverage artefact, not an absence.

Verification

Six independent agents re-derived these claims in parallel, each instructed to refute rather than confirm: on the cancer denominator, an adversarial hunt for a second cancer exercise study, the field-level facts of ST002183, the lung slice, free-text reachability, and the two deposition questions. Outcome: the three headline counts were corrected, the headline

answer survived all four search directions, and eight new findings were added — the crossover control arm, the selective-deposition pattern, the SWHS deposition gap, and the repository defects above. All network access ran through the two allowlisted clients; no synthetic record was substituted for any unavailable one.

Provenance

Item	Value
Corpus listing	/rest/study/study_id/ST/summary — 4,504 rows, 4,503 unique studies, 2026-08-26
Disease vocabulary	/rest/study/study_id/ST/disease — 2,864 rows, 2,729 studies, 258 distinct values
Species, sample source	/rest/study/study_id/ST/species (4,777 rows), /rest/study/study_id/ST/source (4,891 rows)
Per-study factors	/rest/study/study_id/<ST id>/factors — 4,503 of 4,503 retrieved
Keyword search	/rest/study/study_title/<term>/summary — 20 terms, all reachable
Free text	/rest/study/analysis_id/<AN id>/mwtab, prose blocks only
Literature	Europe PMC REST via the allowlisted literature client
Prior snapshot	data/live/mw_corpus_snapshot/manifest.json, 2026-08-24T04:29:22Z

Appendix A — the lung-cancer set (75 studies, source-inclusive construction)

Exercise or physical-activity variable in deposited factors: none, every row.

Study	Species	Source	n	Platform	Released	Title
ST000010	human	Lung	39	LC-MS	2013-05-03	Lung Cancer Cells 4
ST000104	human	Blood	68	NMR	2016-06-18	Factors for Epigenetic Silencing of Lung Cancer Genes
ST000142	human	Lung	4	LC-MS	2015-03-12	H1299 13C-labeled Cell Study
ST000220	human	Lung	21	LC-MS	2016-07-08	Small cell lung cancer metabolome (part II)
ST000367	human	Lung	12	LC-MS	2016-03-21	Distinctly perturbed metabolic networks underlie differential tumor t...
ST000368	human	Blood	192	GC-MS	2016-04-08	Investigation of metabolomic blood biomarkers for detection of adenoc...

Study	Species	Source	n	Platform	Released	Title
ST000369	human	Blood	181	GC-MS	2016-04-03	Investigation of metabolomic blood biomarkers for detection of adenoc...
ST000385	human	Blood	192	GC-MS	2016-04-30	Investigation of metabolomic blood biomarkers for detection of adenoc...
ST000386	human	Blood	180	GC-MS	2016-04-25	Investigation of metabolomic blood biomarkers for detection of adenoc...
ST000388	human	Blood	95	LC-MS	2016-05-01	Serum phosphatidylethanolamine levels distinguish benign from maligna...
ST000389	human	Blood	95	GC-MS	2016-05-01	Serum phosphatidylethanolamine levels distinguish benign from maligna...
ST000390	human	Lung	87	GC-MS	2016-06-18	Metabolomic markers of altered nucleotide metabolism in early stage a...
ST000391	human	Lung	87	LC-MS	2016-06-18	Metabolomic markers of altered nucleotide metabolism in early stage a...
ST000392	human	Blood	183	GC-MS	2016-06-18	Systemic Metabolomic Changes in Blood Samples of Lung Cancer Patients...
ST000396	human	Blood	299	GC-MS	2016-06-18	Lung Cancer Plasma Discovery
ST001269	human	Blood	95	MS(Dir. Inf.)	2019-10-11	Exosomal lipids for classifying early and late stage non-small cell l...
ST001527	human	Tumor cells	72	LC-MS	2022-08-01	Lung cancer metabolomics analysis
ST001610	human	Lung	18	LC-MS	2020-12-09	Control (DMSO 0.1%; v/v) and 10 μ M DRB18 treated A549 lung cancer cel...
ST001779	human	Lung	36	LC-MS	2021-05-21	Untargeted Metabolomics analysis of A549 treated with 0.5 mM extracel...
ST001937	human	Blood	1160	GC-MS/LC-MS	2023-09-11	Comprehensive plasma metabolomics and lipidomics based management of ...

Study	Species	Source	n	Platform	Released	Title
ST002058	mouse	Lung/Muscle	32	LC-MS	2022-02-14	Muscle/Lung/Tumor metabolomics
ST002066	mouse	Lung	31	LC-MS	2022-05-02	Glutaminase inhibition impairs CD8 T cell activation in STK11/Lkb1 de...
ST002082	human	Urine	112	LC-MS	2022-02-24	Predicting dying: a study of the metabolic changes and the dying proc...
ST002120	human	Lung	229	LC-MS	2022-07-20	Feasibility of detecting AC and SCC using UPLC-HRMS based tissue meta...
ST002223	mouse	Heart/Kidney/Liver/Lung	63	LC-MS	2022-08-03	Metabolic profiling of mouse tissues and tissue interstitial fluids
ST002703	human	Lung	12	LC-MS	2023-11-30	Multi-Omics Analysis Revealed a Significant Molecular Changes in Doxo...
ST002719	human	Lung	6	LC-MS	2023-06-21	Comparison of metabolic of A549 cells before and after Gossypol aceta...
ST002755	human	Lung	24	LC-MS	2023-07-18	Metabolomics Profiling of the Antiproliferative, Anti-migratory and A...
ST002773	human	Blood	1152	LC-MS	2024-02-28	A nested case-control study of untargeted plasma metabolomics and lun...
ST002851	mouse	Liver/Lung	30	LC-MS	2023-10-02	Metabolic characterization of liver metastasis organotropism
ST002962	mouse	Lung Tumors	33	LC-MS	2024-01-16	LC/MS detection for NADPH and NADP+ levels in KRAS-driven lung tumors...
ST002963	mouse	Lung Tumors	33	LC-MS	2024-01-16	LC/MS detection for GSH and GSSG levels in KRAS-driven lung tumors wi...
ST002996	human	Lung	230	LC-MS	2023-12-27	Tissue Lipidomic Profiling for Detection of Non-Small Cell Lung Cancer
ST002999	mouse	Cultured cells	56	LC-MS	2023-12-08	Metabolomics and glucose and glutamine labeled isotope tracing analys...

Study	Species	Source	n	Platform	Released	Title
ST003193	human	Liver/Lung/Lymph node/Peritoneal fluid/Spinal cord	26	LC-MS	2024-05-29	Metabolomics study on frozen tissue derived from the tumor and adja...
ST003286	human	Cultured cells	81	NMR	2024-07-03	Interaction between NSCLC cells, CD8+ T cells and immune checkpoint i...
ST003287	human	Cultured cells	22	NMR	2024-07-22	Metabolic profiling and synergistic therapeutic strategies unveil the...
ST003322	human	Breath	108	LC-MS	2024-08-01	EBC metabolomics in non-small cell lung cancer patients and control i...
ST003370	human	Cultured cells	72	LC-MS	2024-08-20	An untargeted metabolomic analysis of acute AFB1 treatment in liver, ...
ST003578	human	Cultured cells	24	FIA-MS	2025-03-31	NRF2 supports non-small cell lung cancer growth independently of CBP/...
ST003581	human	Saliva	1043	MS(Dir. Inf.)	2026-02-01	Rapid and non-invasive early detection of lung cancer by integration ...
ST003599	human	Lung	73	LC-MS	2025-07-25	Multi-omic profiling of squamous cell lung cancer identifies metaboli...
ST003623	human	Cultured cells	32	LC-MS	2025-01-02	NRF2 supports non-small cell lung cancer growth independently of CBP/...
ST003624	human	Cultured cells	16	LC-MS	2025-01-02	NRF2 supports non-small cell lung cancer growth independently of CBP/...
ST003625	human	Cultured cells	60	LC-MS	2025-01-02	NRF2 supports non-small cell lung cancer growth independently of CBP/...
ST003740	human	Blood	115	LC-MS	2026-01-02	Distinct metabolic signatures in non-small cell lung cancer subtype: ...
ST003883	mouse	Blood/Lung/Tumor tissue	219	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...
ST003884	mouse	Tumor cells	108	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...
ST003885	mouse	Culture media	34	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...

Study	Species	Source	n	Platform	Released	Title
ST003886	mouse	Cultured cells	48	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...
ST003887	mouse	Culture media	148	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...
ST003888	mouse	Muscle	51	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...
ST003889	mouse	Lung	49	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...
ST003890	mouse	Liver	51	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...
ST003891	mouse	Blood	51	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...
ST003893	mouse	Tumor tissue	47	LC-MS	2025-05-06	Respiration defects limit serine synthesis required for lung cancer g...
ST003950	human	Blood	78	LC-MS	2025-07-07	Plasma Phospholipids Analysis in Thai EGFR-mutated Non-Small Cell Lun...
ST003989	human	Blood	78	GC-MS	2025-07-07	Plasma Amino Acids Analysis in Thai EGFR-mutated Non-Small Cell Lung ...
ST004109	mouse	Blood	11	LC-MS	2025-09-01	Early adipose tissue wasting in a preclinical model of human lung can...
ST004177	human	Blood	78	GC-MS	2025-09-16	Plasma Organic acids Analysis in Thai EGFR-mutated Non-Small Cell Lun...
ST004182	mouse	Cultured cells	20	LC-MS	2025-09-28	Metabolomics of ASCL1-high and ASCL1-low SCLC cells in mouse models
ST004184	human	Blood	78	GC-MS	2025-10-06	Plasma Sugar and Sugar Alcohols Analysis in Thai EGFR-mutated Non-Sma...
ST004203	mouse	Tumor cells	6	LC-MS	2025-10-10	Cell state-specific metabolic networks govern ferroptosis versus apop...

Study	Species	Source	n	Platform	Released	Title
ST004220	human	Lung Tumors	12	LC-MS	2026-03-23	Untargeted Metabolomics of NCI-H1299 Cells Transfected with siRNA Tar...
ST004221	human	Lung Tumors	13	LC-MS	2026-03-23	Targeted Detection of Cystathionine in NCI-H1299 Cells Transfected wi...
ST004225	human	Lung Tumors	13	LC-MS	2026-03-23	Targeted Detection of Cholesterol, Lanosterol, 7-Dehydrocholesterol a...
ST004253	human	Lung Tumors	13	LC-MS	2026-03-23	Targeted Detection of S-Adenosylmethionine and S-Adenosylhomocysteine...
ST004375	human	Cultured cells	20	LC-MS	2026-03-02	Dihydroorotate dehydrogenase (DHODH) promotes radioresistance in Lung...
ST004447	human	Cultured cells	18	LC-MS	2025-12-22	Determine the impact of serum modulation to lipid homeostasis in a no...
ST004449	human	Cultured cells	12	LC-MS	2025-12-22	Compare lipid homeostasis in two distinct non-small cell lung cancer ...
ST004457	mouse	Lung Tumors	48	LC-MS	2025-12-22	Compare lipid metabolism in lung tumors from mice harboring Kras G12D...
ST004458	human	Cultured cells	12	LC-MS	2025-12-22	Determine the effect of a moderate dose of fumonisin B1 to the lipido...
ST004818	human	Blood	84	LC-MS	2026-05-11	Whole Blood Metabolomics Reveals Coordinated Redox-One-Carbon Axis Di...
ST004975	human	Blood	2268	MS(Dir. Inf.)	2026-07-01	A high-resolution circulating metabolic atlas of lung adenocarcinoma ...
ST005097	mouse	Tumor cells	16	LC-MS	2026-08-16	Dietary α -Ketoglutarate Modulates Lung Adenocarcinoma Growth and Epig...

Appendix B — every cancer study matching an exercise-side keyword, adjudicated

Study	MW disease	Matched term	Verdict
ST000385	Cancer	training	Homonym — “training set” is a machine-learning split
ST001005	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001006	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001009	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001010	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001011	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001012	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001013	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001014	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001015	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001016	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001017	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001018	Cachexia/Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST001467	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002183	Multiple myeloma	exercise	Genuine — randomised exercise vs waitlist arms
ST002824	Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002842	Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002843	Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002881	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002882	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002883	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure

Study	MW disease	Matched term	Verdict
ST002884	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002885	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002886	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002887	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002888	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002889	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST002904	Cachexia	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST003919	Cancer/Colitis	aerobic	Homonym — keyword match on unrelated text
ST004109	Cancer	cachexia	Cachexia — wasting phenotype, not an activity exposure
ST004582	Cancer	fitness	Homonym — “fitness” means cellular fitness under hypoxia

False positives found independently by a verification agent over the same set and confirmed here: [ST002010](#) (Chemoresistance Status:Resistant), [ST004922](#) (Palbociclib resistance:Sensitive), [ST000248](#) (Cell population:Fast-cycling cells).

Appendix C — every prose hit in the 798-study cancer prose sweep, adjudicated

Study	MW disease	Term	Verdict
ST002183	Multiple myeloma	exercise, cardiorespiratory	Genuine — randomised exercise intervention
ST004161	Cancer	physical function	Genuine — handgrip strength, stair-climb power, CT muscle CSA; not deposited
ST004166	Cancer	physical function	Genuine — same measures, muscle biopsy arm; not deposited
ST002163	Cancer	wheel-running	Genuine — voluntary wheel-running activity in tumour-bearing mice; not deposited
ST002167	Cancer	wheel-running	Shared abstract — this arm is AML12 cell culture, not the mouse study
ST003315	Cancer	ECOG	Eligibility criterion (ECOG PS ≤ 2), not a measured variable

Study	MW disease	Term	Verdict
ST001237	Cancer	performance status	Embedded in the composite CRF_MSKCC_Risk_Group, not deposited separately

Appendix D — all 35 corpus-wide exercise-factor studies

Context for the headline: these are every study in MW, in any disease area, whose deposited factors mention an exercise term. Exactly one is a cancer study.

Study	MW disease	Matched term	Title
ST000161	not annotated	exercis	Plasma Nucleotide/adenosine concentrations (Human AxP Batch 3)
ST000283	not annotated	exercis	Plasma Nucleotide/adenosine concentrations (Human AxP Batch 4)
ST000315	Obesity	exercis	Metabolomics and Childhood Obesity: A Pilot and Feasibility Stu...
ST000387	not annotated	exercis	Changes in the metabalome and lipidome in response to exercise ...
ST000640	Diabetes	exercis	Targeted NEFA in American Indian Adolescents (part I)
ST000641	Diabetes	exercis	Targeted Amino Acids in American Indian Adolescents (part II)
ST000645	Muscular dystrophy	sedentary	Effects of Exercise on Dystrophic Mouse Muscle Amino Acids (par...
ST000646	Muscular dystrophy	sedentary	Effects of Exercise on Dystrophic Mouse Muscle TCA Cycle (part ...
ST000647	Muscular dystrophy	sedentary	Effects of Exercise on Dystrophic Mouse Muscle Non-Esterified F...
ST000648	Muscular dystrophy	sedentary	Effects of NO Donor Therapy on the Dystrophic Mouse Muscle Amin...
ST000649	Muscular dystrophy	sedentary	Effects of NO Donor Therapy on the Dystrophic Mouse Muscle Non-...
ST000650	Muscular dystrophy	sedentary	Effects of NO Donor Therapy on the Dystrophic Mouse Muscle TCA ...
ST000671	not annotated	exercis	LCR/HCR rat mitochondrial study

Study	MW disease	Matched term	Title
ST000710	not annotated	sedentary	Influence of exercise and antibiotic on fecal SCFA
ST000762	Scleroderma	exercis	Plasma Nucleotide/adenosine concentrations in patients with scler...
ST000763	Scleroderma	exercis	Untargeted metabolomics and lipidomics of scleroderma PAH disco...
ST000777	Obesity	exercis	Metabolic Adaptations to Chronic and Acute Exercise in Overweig...
ST000897	Heart injury	exercis, sedentary	Untargeted metabolomics analysis of ischemia-reperfusion injure...
ST001068	not annotated	treadmill	The proteomic and metabolomic characterization of exercise-indu...
ST001070	not annotated	treadmill	The global proteomic characterization of exercise-induced sweat...
ST001251	Endotoxemia	trained	The effects of a training program encompassing cold exposure, b...
ST001492	not annotated	immobiliz	Global metabolomics of IFN γ induced neurogenic NSCs seeded on hydr...
ST001745	not annotated	trained	Metabolomic profiling of the rat hippocampus across development...
ST001749	Alzheimers disease	physical activity	REACH Metabolomics Study
ST002081	not annotated	exercis	Dynamic Lipidome Alterations Associated with Human Health, Dise...
ST002183	Multiple myeloma	exercis	Individualized exercise intervention for people with multiple m... ← the cancer study
ST002931	not annotated	sedentary	Analyzing Metabolic Alterations in the Gut, Blood, and Brain of...
ST002938	Atrial fibrillation	athlet	Role of PI3K in Atrial Myopathy: Insights from Transgenic Mouse...
ST003686	Cardiovascular disease	exercis	Characterization of the transpulmonary metabolome at the inters...
ST003721	not annotated	immobiliz	Metabolomic analysis of skeletal muscle from control, immobiliz...

Study	MW disease	Matched term	Title
ST003722	Muscle atrophy	immobiliz	Metabolomic analysis of plasma from control, immobilized and Fo...
ST003724	Muscle atrophy	immobiliz	Metabolomic analysis of skeletal muscle from control WT, immobi...
ST003807	Diabetes	exercis	A lipidomic exploration of the effects of high-intensity interv...
ST004061	not annotated	trained	In Vitro Method Demonstrating Trained Immunity as a Distinctive...
ST004721	not annotated	exercis	High relevance of fatty acid oxidation in a migrating mammal, t...